

10.5.1

EE25BTECH11033 - Kavin

Question:

Let ABC be a right triangle in which $AB = 6\text{cm}$, $BC = 8\text{cm}$ and $\angle B = 90^\circ$. BD is the perpendicular from B on AC . The circle through B, C, D is drawn. Construct the tangents from A to this circle.

Solution:

Let,

$$\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 0 \\ 6 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} 8 \\ 0 \end{pmatrix} \quad (1)$$

Let the equation of the line passing through A and C be,

$$\mathbf{n}^\top \mathbf{x} = c \quad (2)$$

The direction vector of line AC is given by,

$$\mathbf{m} = \mathbf{A} - \mathbf{C} \equiv \begin{pmatrix} -4 \\ 3 \end{pmatrix} \quad (3)$$

Therefore, the normal vector of the desired line is given by,

$$\mathbf{n} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \quad (4)$$

$$c = \mathbf{n}^\top \mathbf{A} = \begin{pmatrix} 3 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 6 \end{pmatrix} = 24 \quad (5)$$

Let D be the foot of the perpendicular from B to the line

$$\begin{pmatrix} \mathbf{m} & \mathbf{n} \end{pmatrix}^\top \mathbf{D} = \begin{pmatrix} \mathbf{m}^\top \mathbf{B} \\ c \end{pmatrix} \quad (6)$$

$$\begin{pmatrix} -4 & 3 \\ 3 & 4 \end{pmatrix} \mathbf{D} = \begin{pmatrix} \mathbf{m}^\top \mathbf{B} \\ 24 \end{pmatrix} \quad (7)$$

$$\mathbf{m}^\top \mathbf{B} = \begin{pmatrix} -4 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0 \quad (8)$$

$$\Rightarrow \begin{pmatrix} -4 & 3 \\ 3 & 4 \end{pmatrix} \mathbf{D} = \begin{pmatrix} 0 \\ 24 \end{pmatrix} \quad (9)$$

The augmented matrix is given by,

$$\left(\begin{array}{cc|c} -4 & 3 & 0 \\ 3 & 3 & 24 \end{array} \right) \quad (10)$$

$$R_1 \rightarrow \frac{-1}{4}R_1 \Rightarrow \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 3 & 3 & 24 \end{array} \right) \quad (11)$$

$$R_2 \rightarrow R_2 - 3R_1 \Rightarrow \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 0 & 25/4 & 24 \end{array} \right) \quad (12)$$

$$R_2 \rightarrow \frac{4}{25}R_2 \Rightarrow \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 0 & 1 & 96/25 \end{array} \right) \quad (13)$$

$$R_1 \rightarrow R_1 + \frac{3}{4}R_2 \Rightarrow \left(\begin{array}{cc|c} 1 & 0 & 72/25 \\ 0 & 1 & 96/25 \end{array} \right) \quad (14)$$

$$\mathbf{D} = \begin{pmatrix} 72/25 \\ 96/25 \end{pmatrix} \quad (15)$$

To find the circle through $\mathbf{B}, \mathbf{C}, \mathbf{D}$ the parameters are given by

$$\begin{pmatrix} 2\mathbf{B} & 2\mathbf{C} & 2\mathbf{D} \\ 1 & 1 & 1 \end{pmatrix}^\top \begin{pmatrix} \mathbf{u} \\ f \end{pmatrix} = - \begin{pmatrix} \|\mathbf{B}\|^2 \\ \|\mathbf{C}\|^2 \\ \|\mathbf{D}\|^2 \end{pmatrix} \quad (16)$$

$$\begin{pmatrix} 0 & 0 & 1 \\ 16 & 0 & 1 \\ 144/25 & 192/25 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ f \end{pmatrix} = \begin{pmatrix} 0 \\ -64 \\ -576/25 \end{pmatrix} \quad (17)$$

The augmented matrix is given by,

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 16 & 0 & 1 & -64 \\ 144/25 & 192/25 & 1 & -576/25 \end{array} \right) \quad (18)$$

$$R_1 \leftrightarrow R_2 \Rightarrow \left(\begin{array}{ccc|c} 16 & 0 & 1 & -64 \\ 0 & 0 & 1 & 0 \\ 144/25 & 192/25 & 1 & -576/25 \end{array} \right) \quad (19)$$

$$R_1 \rightarrow \frac{1}{16}R_1 \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 0 & 1 & 0 \\ 144/25 & 192/25 & 1 & -576/25 \end{array} \right) \quad (20)$$

$$R_3 \rightarrow R_3 - \frac{144}{25}R_1 \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 192/25 & 16/25 & -504/25 \end{array} \right) \quad (21)$$

$$R_2 \leftrightarrow R_3 \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 192/25 & 16/25 & -504/25 \\ 0 & 0 & 1 & 0 \end{array} \right) \quad (22)$$

$$R_2 \rightarrow \frac{25}{192}R_2 \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 1 & 1/12 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right) \quad (23)$$

$$R_2 \rightarrow R_2 - \frac{1}{12}R_3, \quad R_1 \rightarrow R_1 - \frac{1}{16}R_3 \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right) \quad (24)$$

$$\Rightarrow \mathbf{u} = \begin{pmatrix} -4 \\ 0 \end{pmatrix} \text{ and } f = 0 \quad (25)$$

Let $\mathbf{n}_1^\top \mathbf{x} = c_1$ and $\mathbf{n}_2^\top \mathbf{x} = c_2$ be the equation of tangents to the circle.
The external point from which the tangents are drawn is $\mathbf{A}$
Compute Σ

$$\Sigma = (\mathbf{V}\mathbf{h} + \mathbf{u})(\mathbf{V}\mathbf{h} + \mathbf{u})^\top - g(\mathbf{h})\mathbf{V} \quad (26)$$

for $\mathbf{h} = \mathbf{A}$

$$g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V}\mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f = \|\mathbf{h}\|^2 + f = 36 + 0 = 36 \quad (27)$$

$$\Sigma = (\mathbf{V}\mathbf{h} + \mathbf{u})(\mathbf{V}\mathbf{h} + \mathbf{u})^\top - g(\mathbf{h})\mathbf{V} = \begin{pmatrix} -4 \\ 6 \end{pmatrix} \begin{pmatrix} -4 & 6 \end{pmatrix} - 36 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (28)$$

$$\Rightarrow \Sigma = \begin{pmatrix} -20 & -24 \\ -24 & 0 \end{pmatrix} \quad (29)$$

The eigenvalues of the matrix Σ are given by

$$|\Sigma - \lambda \mathbf{I}| = 0 \quad (30)$$

$$\Rightarrow \begin{vmatrix} -20 - \lambda & -24 \\ -24 & -\lambda \end{vmatrix} \quad (31)$$

$$\Rightarrow (-20 - \lambda)(-\lambda) - (-24)(-24) = 0 \quad (32)$$

$$\Rightarrow \lambda^2 + 20\lambda - 576 = 0 \quad (33)$$

$$\implies \lambda_1 = 16 \text{ and } \lambda_2 = -36 \quad (34)$$

The eigenvectors of the matrix Σ are,

$$\mathbf{p}_1 = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \text{ and } \mathbf{p}_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad (35)$$

Define,

$$\mathbf{P} = \left(\frac{\mathbf{p}_1}{\|\mathbf{p}_1\|} \quad \frac{\mathbf{p}_2}{\|\mathbf{p}_2\|} \right) = \begin{pmatrix} \frac{2}{\sqrt{13}} & \frac{3}{\sqrt{13}} \\ \frac{-3}{\sqrt{13}} & \frac{2}{\sqrt{13}} \end{pmatrix} \quad (36)$$

The direction vectors of the two tangents are,

$$\mathbf{m} = \mathbf{P} \begin{pmatrix} \sqrt{|\lambda_2|} \\ \pm \sqrt{|\lambda_1|} \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{13}} & \frac{3}{\sqrt{13}} \\ \frac{-3}{\sqrt{13}} & \frac{2}{\sqrt{13}} \end{pmatrix} \begin{pmatrix} 6 \\ \pm 4 \end{pmatrix} \quad (37)$$

$$\implies \mathbf{m}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ and } \mathbf{m}_2 = \begin{pmatrix} 12 \\ -5 \end{pmatrix} \quad (38)$$

$$\implies \mathbf{n}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \mathbf{n}_2 = \begin{pmatrix} 5 \\ 12 \end{pmatrix} \quad (39)$$

Also,

$$c_1 = \mathbf{n}_1^\top \mathbf{A} \text{ and } c_2 = \mathbf{n}_2^\top \mathbf{A} \quad (40)$$

$$c_1 = 0 \text{ and } c_2 = 72 \quad (41)$$

Therefore, the line $\begin{pmatrix} 5 & 12 \end{pmatrix} \mathbf{x} = 72$ and the $\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 0$ are the two tangents.

